STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 2: Motion in a Straight Line

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Accept equivalent correct methods.

Suggested mark allocation:

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

Q1 - 1 mark

v=dx/dt.

Q2 - 1 mark

Velocity.

Q3 - 1 mark

Zero.

Q4 - 1 mark

No.

Q5 - 1 mark

m s^-2.

Q6 - 1 mark

Zero.

Q7 - 1 mark

v=u+at.

Q8 - 1 mark

Signed displacement.

Q9 - 2 marks

5 m/s.

Q10 - 2 marks

2 m/s^2.

Q11 - 2 marks

5 s.

Q12 - 2 marks

25 m/s.

Q13 - 2 marks

11 m; 5 m east.

Q14 - 2 marks

12 m/s.

Q15 - 3 marks

v=16 m/s; s=55 m.

Q16 - 3 marks

a=-4 m/s^2; s=50 m.

Q17 - 3 marks

a=2 m/s^2; displacement=48 m.

Q18 - 3 marks

v=24 m/s; s=144 m.

Q19 - 4 marks

5 s; 62.5 m; acceleration opposes motion.

Q20 - 4 marks

v=3-2t; a=-2 m/s^2; t=1.5 s.